THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

(NMTC- at ***SUB-JUNIOR LEVEL*** *VII & VIII Standards)*

Saturday, 23rd August 2014.

Note:

1) *Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.*
2) *Diagrams are only visual aids; they are not drawn to scale.*
3) *You are free to do rough work on separate sheets.*
4) *Duration of the test: 2 p.m. to 4. p.m.- 2 hours.*

PART – A

Note:

- *Only one of the choices A, B, C, D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
- *For each correct response you get 1 mark;* ***for each incorrect response you lose ½ mark.***

1. If 150% of a certain number is 300, then 30% of the number is

   a) 50 b) 60 c) 70 d) 65

2. The sum of five distinct positive integers is 90. What can be the second largest number of the five at most?

   a) 82 b) 43 c) 34 d) 73

3. The length of two sides of an isosceles triangle are 5 units and 16 units. The perimeter of the triangle (in the same units) is

   a) 26 b) 37 c) 26 or 37 d) none of these

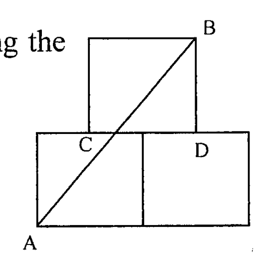
4. $ABCD$ is a rectangle. P is the mid-point of DC and Q is a point on AB such that $AQ = \frac{1}{3}$ AB. What fraction of the area of $ABCD$ is $AQPD$.

   a) $\frac{1}{2}$ b) $\frac{3}{4}$ c) $\frac{2}{7}$ d) $\frac{5}{12}$

5. The number of digits when $(999999999999)^2$ is expanded is

   a) 26 b) 24 c) 32 d) 16

6. Three equal squares are kept as in the diagram. C, D being the mid points of the respective sides of the lower squares. If $AB = 100$ cm, Area of each square is (in cm^2)

   a) 1200 b) 1500

   c) 900 d) 1600

7. Samrud wrote 4 different natural numbers. He chose three numbers at a time and added them each time. He got the sums as 115, 153, 169, 181. The largest of the numbers Samrud first wrote is

a) 37 b) 48 c) 57 d) 91

8. Saket wrote a two digit number. He added 5 to the tens digit and subtracted 3 from the units digit of the number. The resulting number is twice the original number. The original number is

a) 47 b) 74 c) 37 d) 73

9. Five consecutive natural numbers cannot add up to

a) 225 b) 222 c) 220 d) 200

10. In the adjoining figure the different numbers denote the area of the corresponding rectangle in which the number is there. The value of x is

2014	1007
x	125

a) 3014 b) 1125 c) 2139 d) 250

11. What is the remainder when $2^{87}+3$ is divided by 7

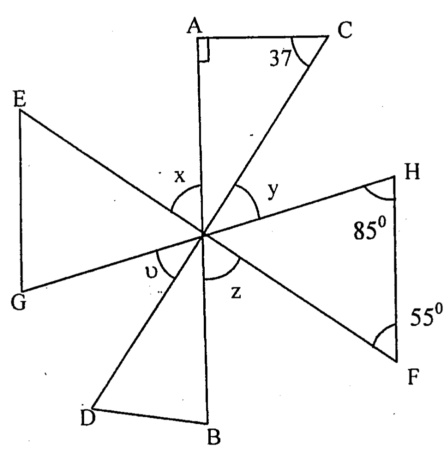
a) 2 b) 3 c) 4 d) 5

12. If Mahadevan gets 71 in his next examination, his average will be 83. If the gets 99, his average will be 87. How many exams Mahadevan has already taken?

a) 3 b) 4 c) 5 d) 6

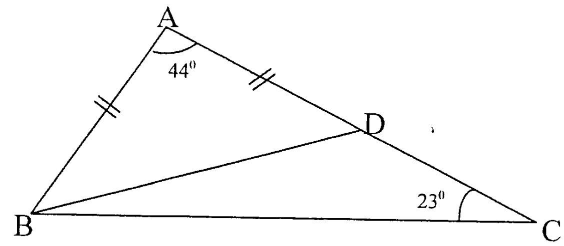
13. In the adjoining figure, AB, CD, EF and GH are straight lines passing through a single point. The value of

$\angle x + \angle y + \angle z + \angle \upsilon =$



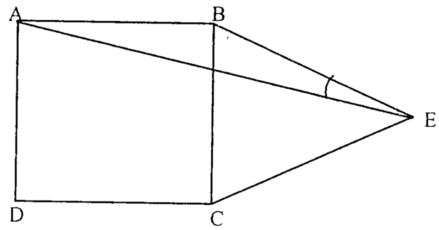
a) 155^o b) 164^o c) 174^o d) 148^o

14. In the adjoining diagram $AB = AD$. $\angle DCB = 23^0$. The measure of $\angle DBC$ is



a) 55^o b) 58^o c) 56^o d) 45^o

15. In the adjoining figure $ABCD$ is a square. BCE is an equilateral triangle. The measure of $\angle BEA$ is



a) 15^o b) 20^o c) 18^o d) 16^o

PART – B

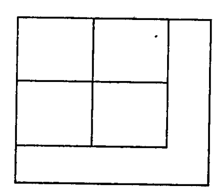
Note:

- *Write the correct answer in the space provided in the response sheet.*
- *For each correct response you get I mark;* ***for each incorrect response you lose ¼ mark.***

16. The smallest multiple of 9 with no odd digits is ___________.

17. The sum of the squares of the length of the three sides of a right triangle is 800. The length of the hypotenuse is______________.

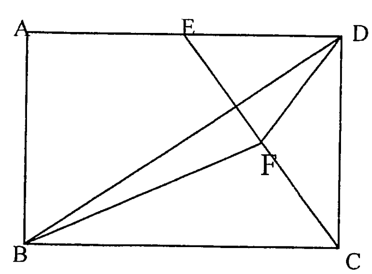
18. In the adjoining figure, the biggest square of area 125 cm^2 is divided into 5 equal parts of same area. (4 squares and the L shaped figure).
The shorter side of the L shape is $k(\sqrt{5}-2)$.
Where k is a natural number. Then k = ___________ .



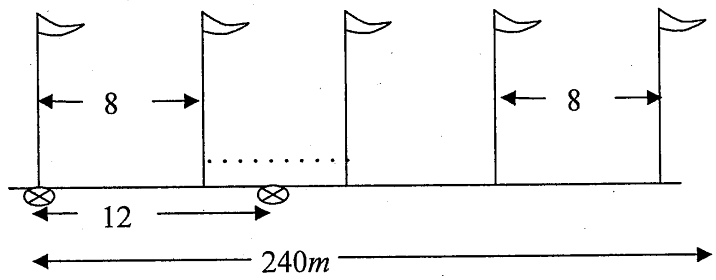
19. Five years ago, the average age of A, B, C and D is 45 years. E joins them now. The average age of all the five now is 49 years. The present age of E is __________.

20. In the sequence 1, 1, 1, 2, 1, 3, 1, 4, 1, 5 . . . The 2014th term is _____________.

21. In the adjoining figure $ABCD$ is a rectangle. E is the midpoint of AD. F is the midpoint of EC. The area of the rectangle $ABCD$ is $120\,cm^2$.
The area of the triangle BDF is ___________ cm^2



22. Flag poles are installed along a road side on every 8 metres. Markings are made every $12m$ as shown. The length of the road is $240m$. The number of markings made beside the flag poles are ________________.



23. The number of different natural numbers 'n' for which $n^2 - 440$ is a perfect square is ______.

24. Aruna, Bhavani and Christina wear a saree of different colours (blue, yellow or green). It is known that, if Aruna wears blue then Bhavani wears green, if Aruna wears yellow Christina wears green and if Bhavani does not wear yellow, Christina wears blue. The saree Aruna is wearing has the colour _____________.

25. Arish found the value of 3^{19} to be $11a2261467$. He found all the digits correctly except the digit denoted by a. The value of 'a' is ________.

26. The daily wages of two persons are in the ratio 3: 5. They work in a place and the employer is satisfied with their work and gives Rs 20 more to each. Then the ratio of their wages comes to 13: 21. The sum of the original wages of the two persons is __________.

27. The value of x which satisfies the equation $\dfrac{5}{6-\dfrac{5}{6-\dfrac{5}{6-x}}} = 1$ is _______________.

28. The age of a father is three times that of his son 10 years after the fathers age is twice that of his son . The fathers age will be 60 after _________years.

29. The number of (x, y, z) which satisfies the equations $xy = 6$, $yz = 15$, $zx = 10$ simultaneously is ______________.

30. If $\dfrac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}} = \dfrac{1}{2}$. The value of $\dfrac{a^2+ab+b^2}{a^2-ab+b^2}$ in the form $\dfrac{p}{q}$ is ____________.

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